

SR-IOV process under WS

Test Case 目的

在 Windows OS 內使用 LAN vEthernet SR-IOV 功能

Test Case 基本欄位定義

Title 、 Steps 、 Note 、 Tags

Test Case 撰寫

Title : SR-IOV process under WS

Note :

1. 在進入 OS 前必須先進 BIOS 確認要使用 SR-IOV 功能的網卡是否已啟用 SR-IOV 相關選項, "以下選項"數值也務必確認是"預設值"(Broadcom 網卡有回復預設值選項)。

1-1. Number of VFs Per PF

1-2. Number of MSI-X Vectors per VF

1-3. Maximum Number of PF MSI-X vectors

2. 進入 Windos OS 後,確認 Device Manager/你要啟用 SR-IOV 的 LAN card port/Advanced/SR-IOV 與 Virtual Machine Queues 兩個選項數值是 Enabled。

3. 如果在 Hyper-V Manager 內已經做好所有 SR-IOV 啟用設定後開啟 VM OS, VM OS 的網卡狀態是"Degraded"而不是"status OK, (SR-IOV active)"的話, 請進入 VM OS 內上 driver。

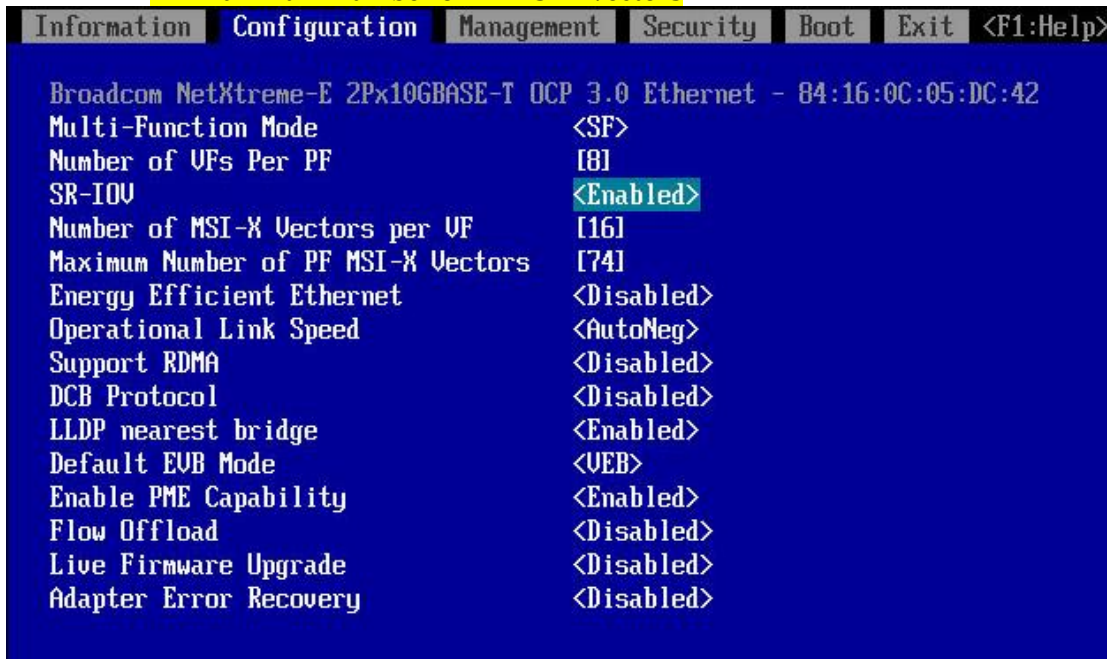
Steps :

1. BIOS/N210TP OCP/Device Configuration Menu

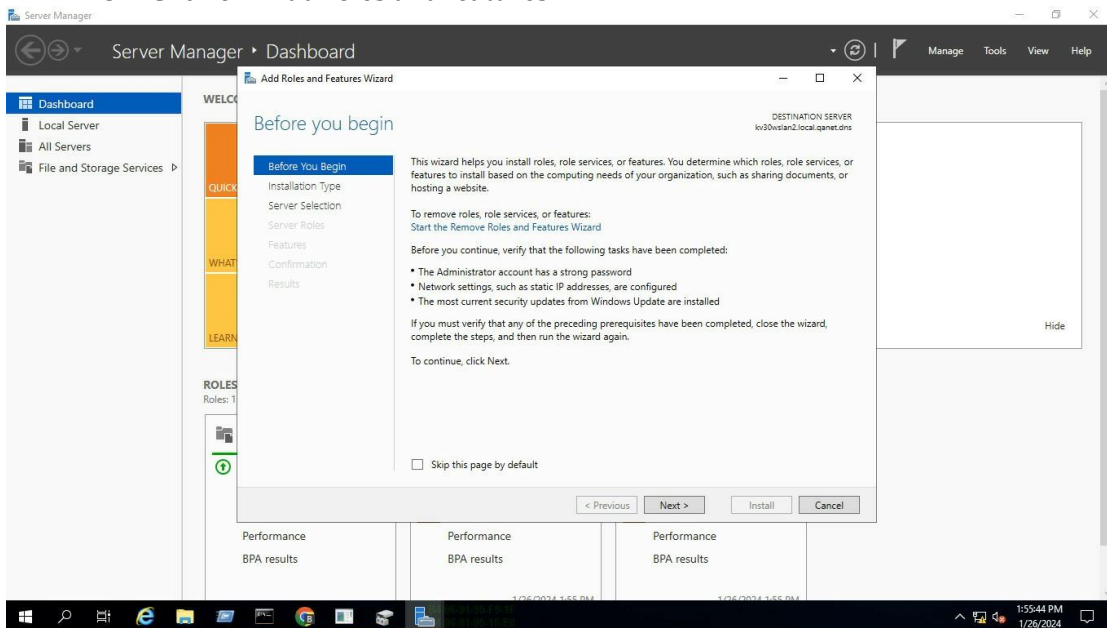


2. BIOS/N210TP OCP/Device Configuration Menu/SR-IOV Enabled
★ Make sure the following BIOS options are set to default values.

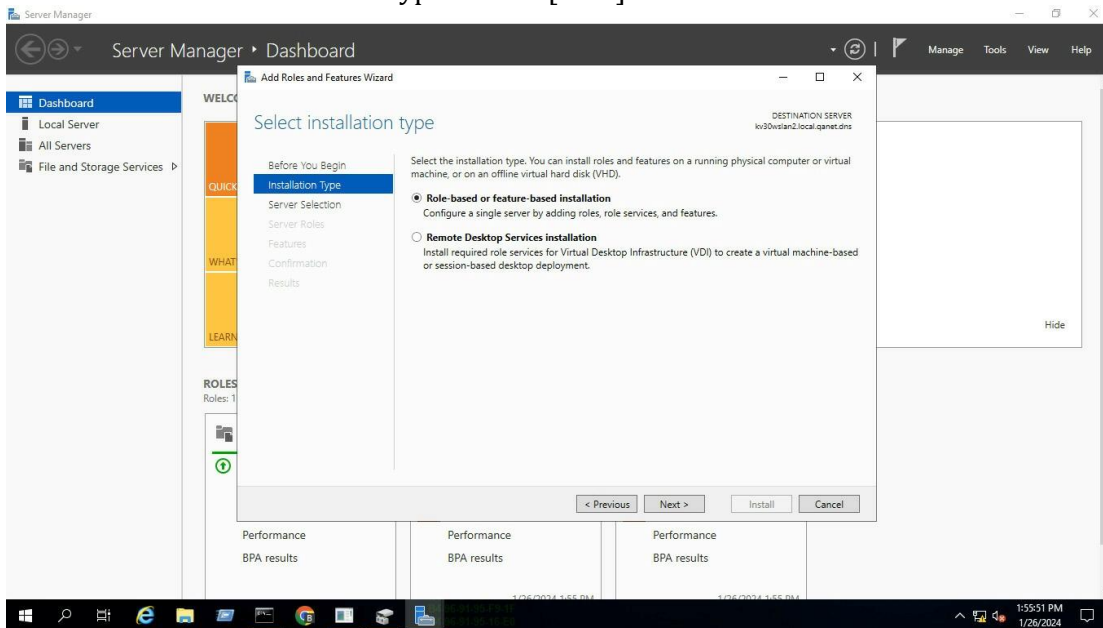
- I. Number of VFs Per PF
- II. Number of MSI-X Vectors per VF
- III. Maximum Number of PF MSI-X vectors



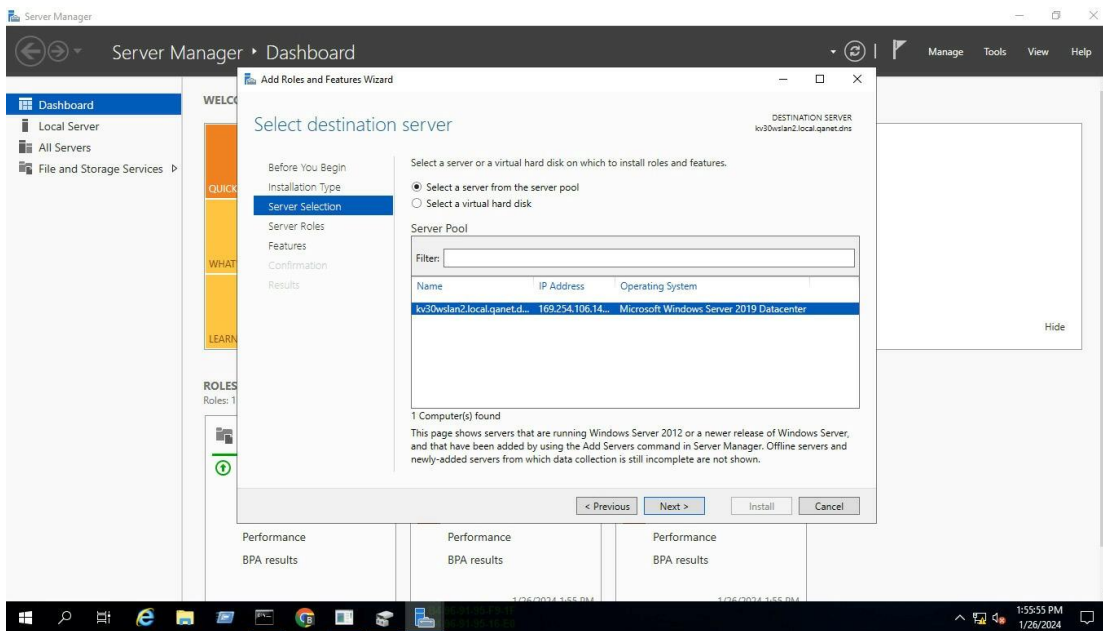
3. Click on “Add roles and features.”



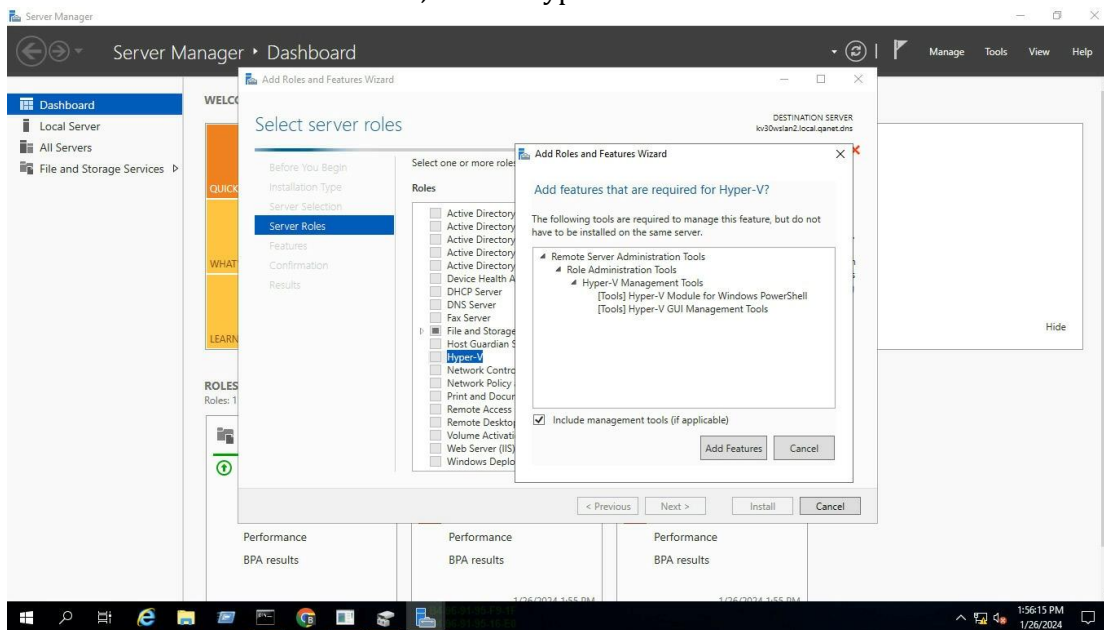
4. “Select installation type” – click [Next]



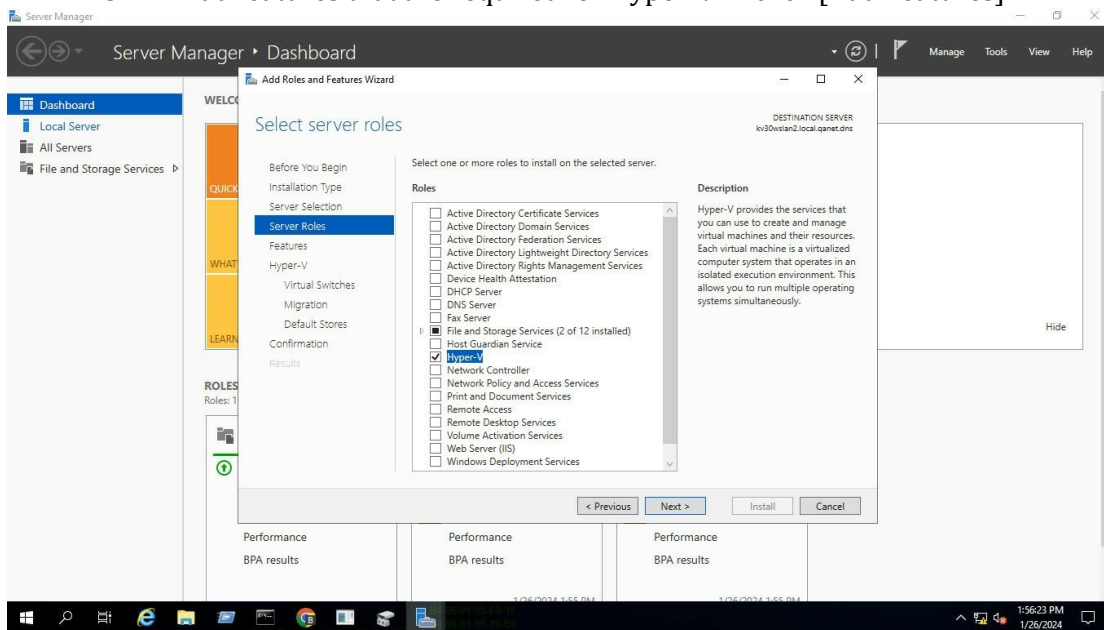
1. “Select destination server”, make sure local server is selected – click [Next]



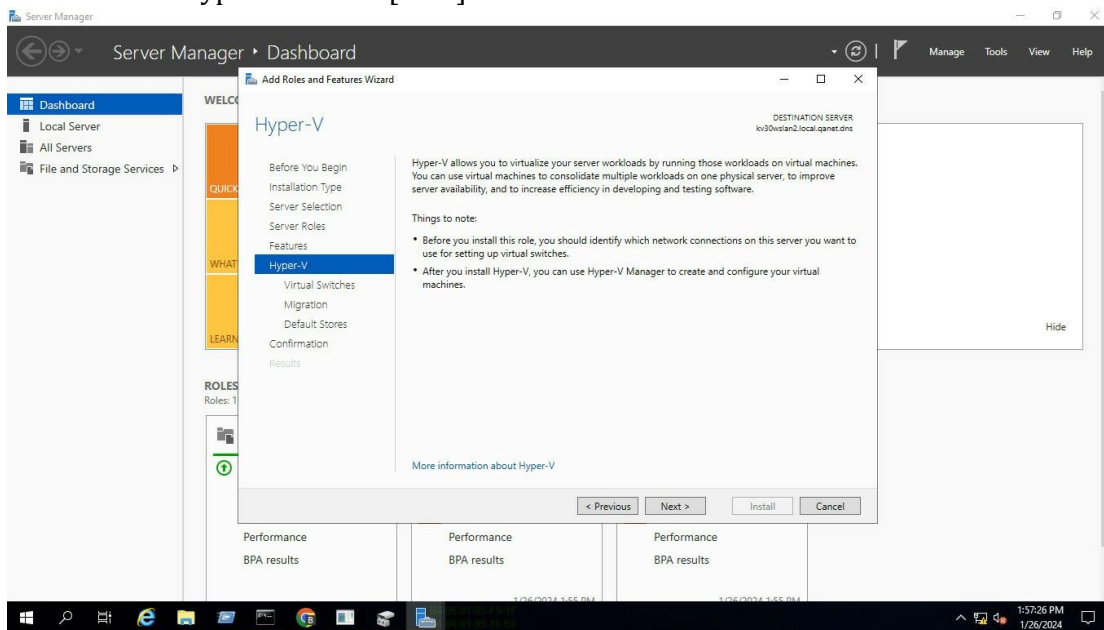
2. “Select server roles”, select “Hyper-V”



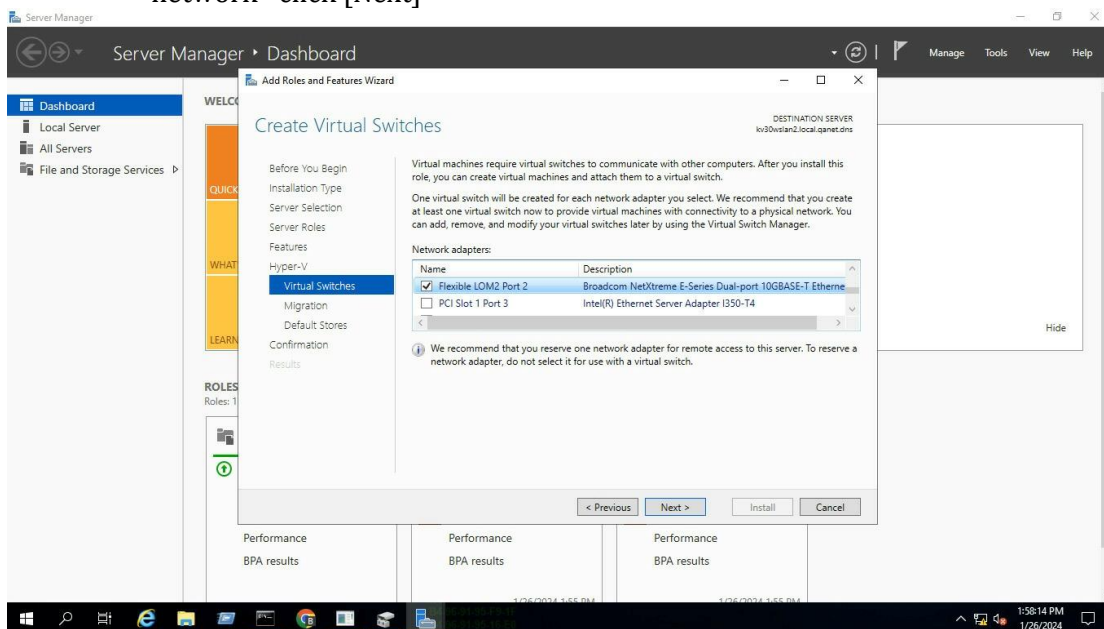
3. “Add features that are required for Hyper-V” – click [Add Features]



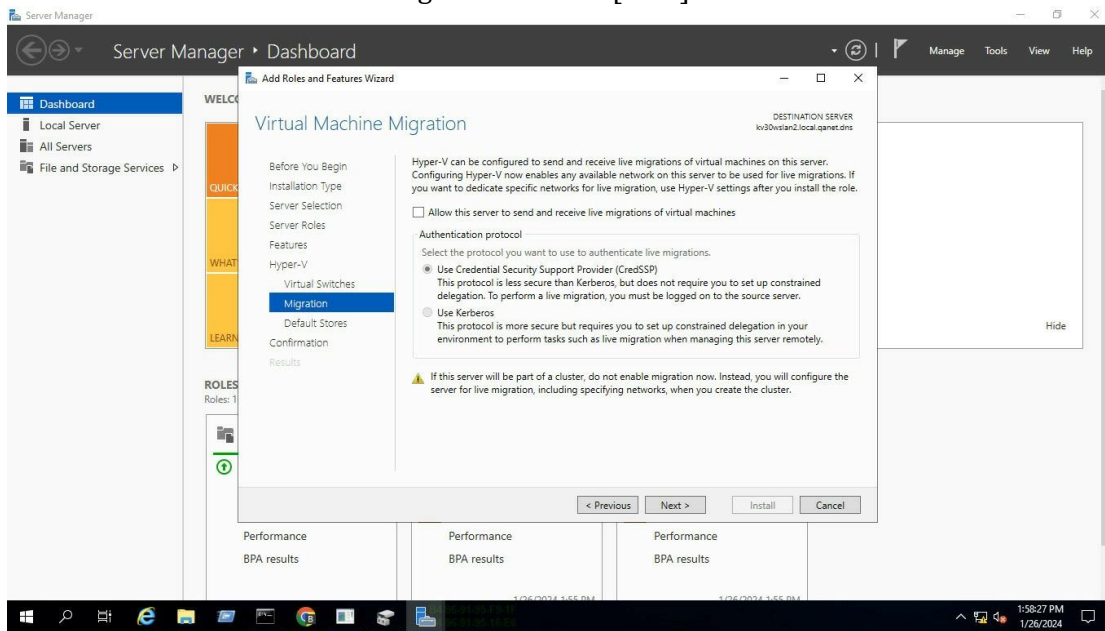
4. “Hyper-V” – click [Next]



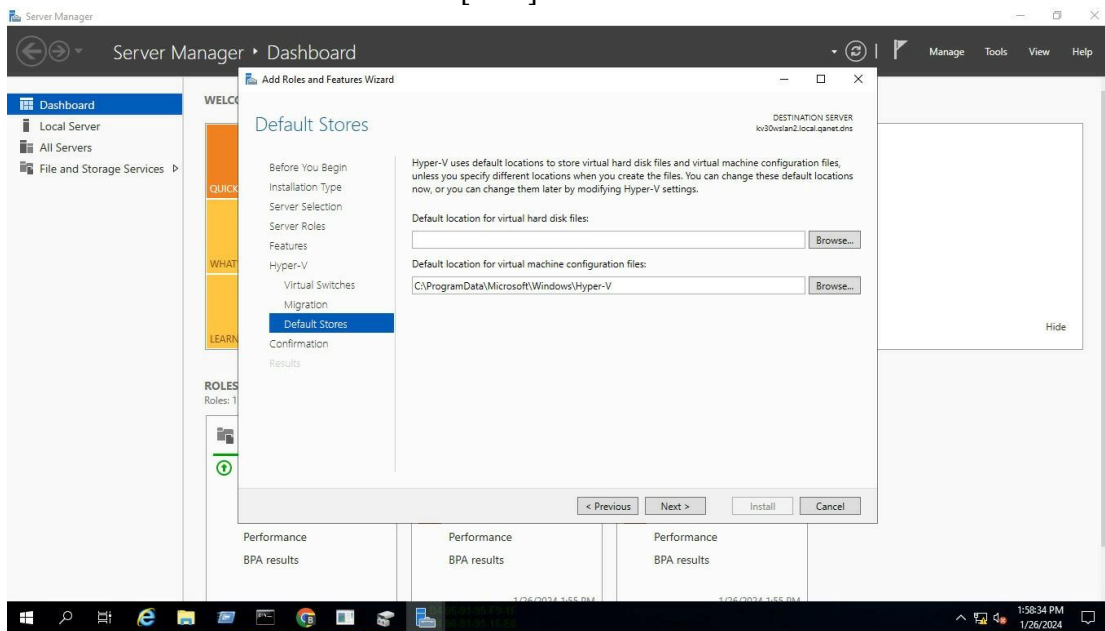
5. “Create virtual switches”, select LAN controller with connection to lab network– click [Next]



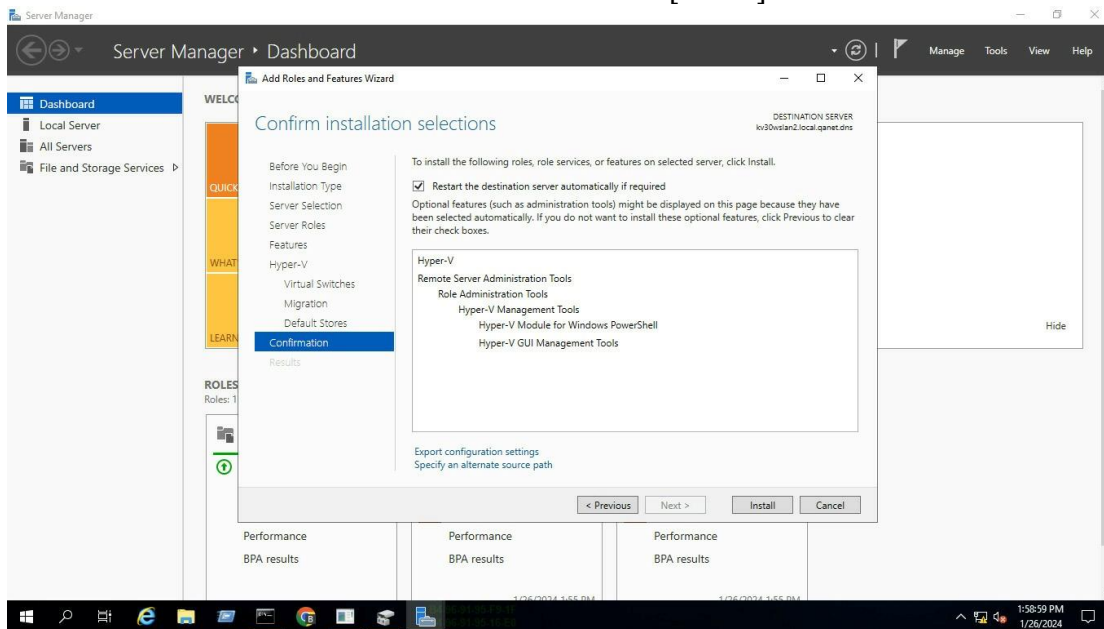
6. “Virtual Machine Migration” – click [Next]



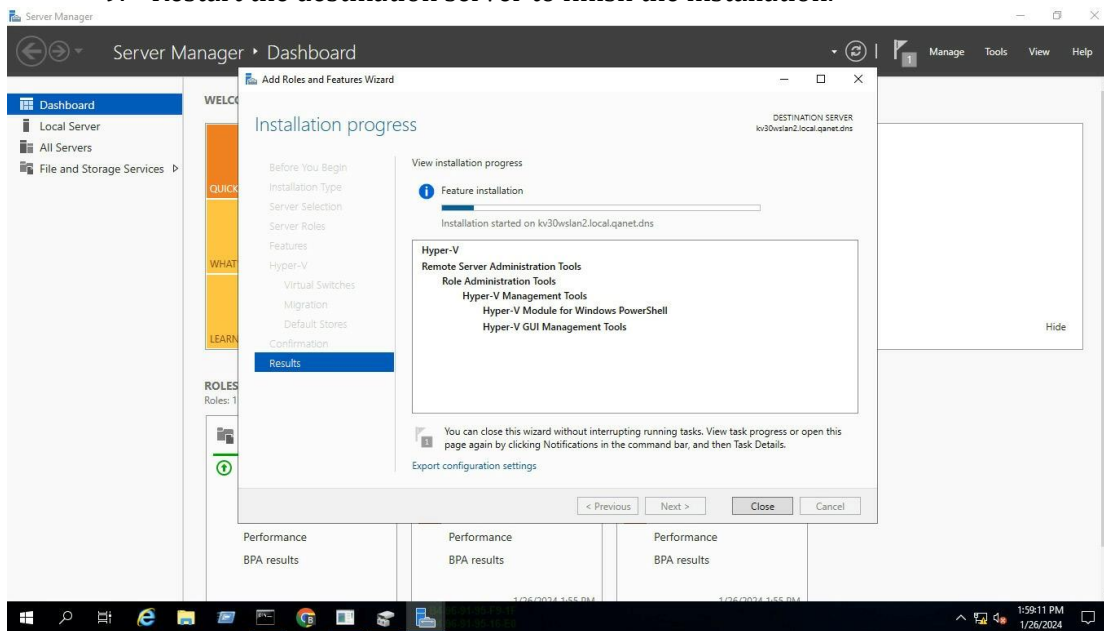
7. “Default Stores” – click [Next]



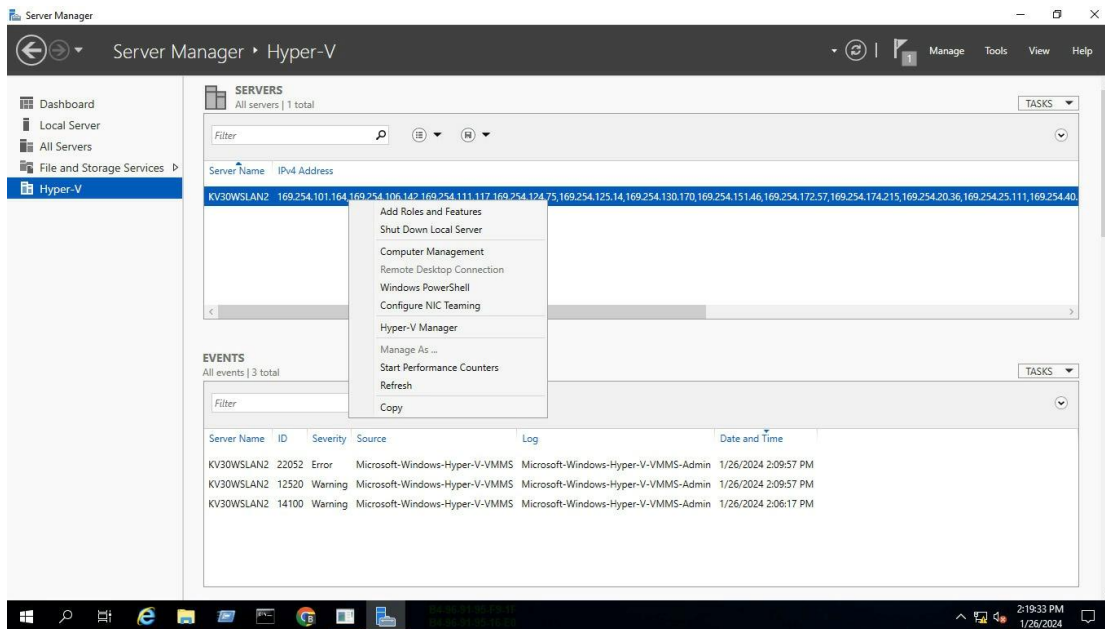
8. “Confirm installation selections” – Click [Install]



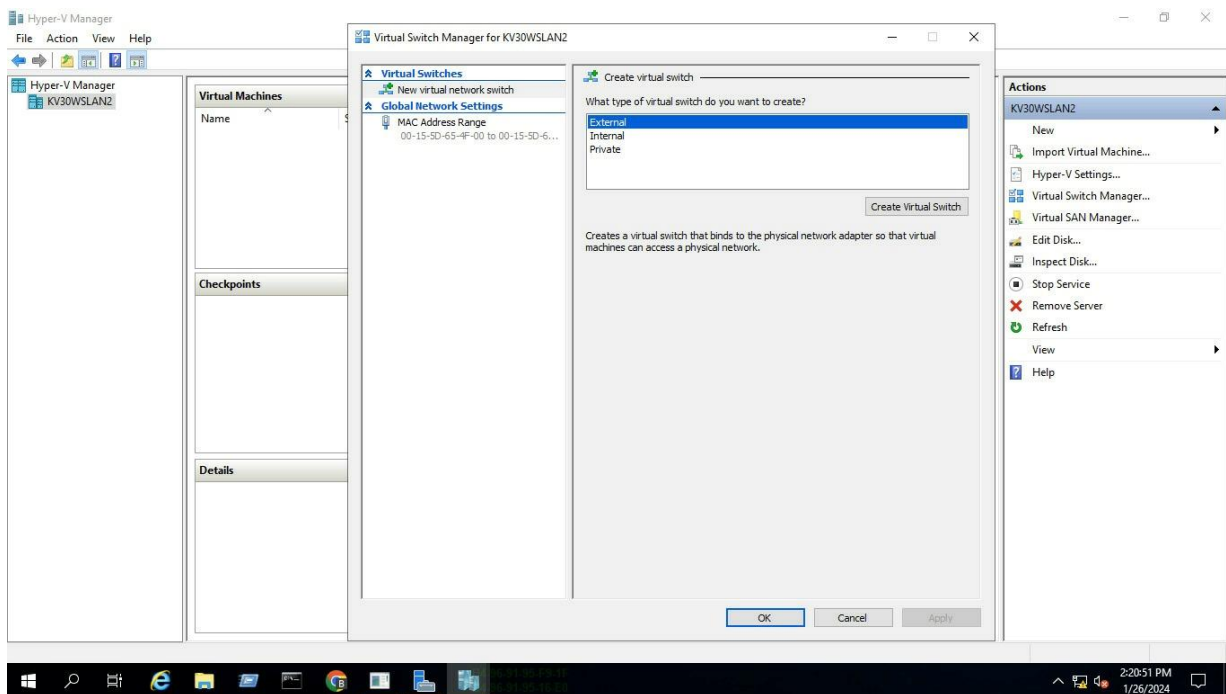
9. Restart the destination server to finish the installation.



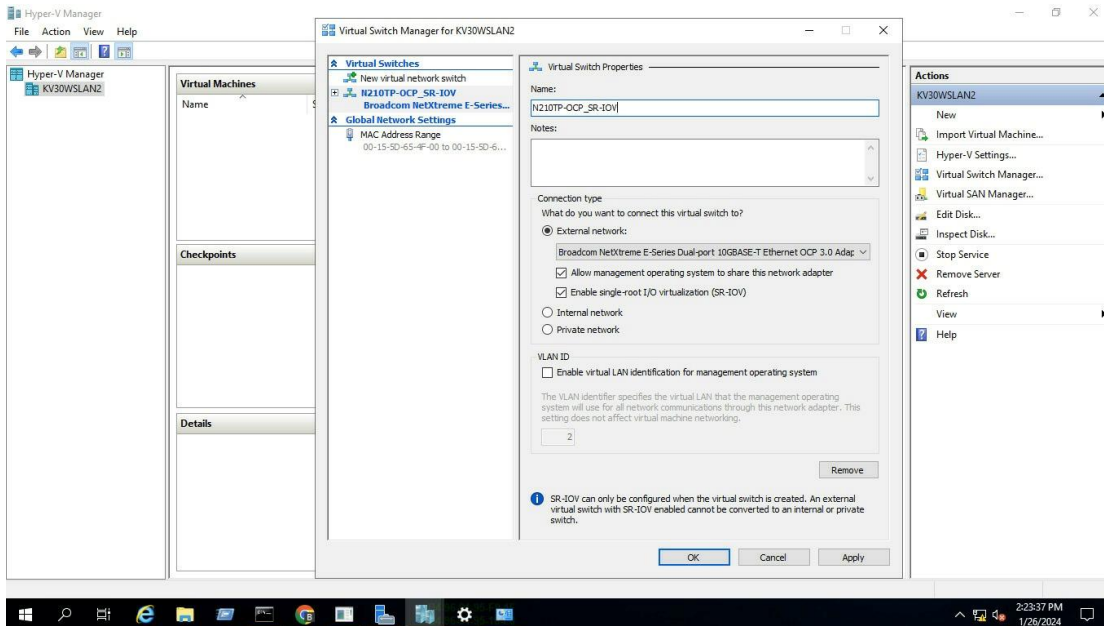
10. Open Hyper-V Manager.



11. “New virtual network switch” and “External” is selected. Click on [Create Virtual Switch]

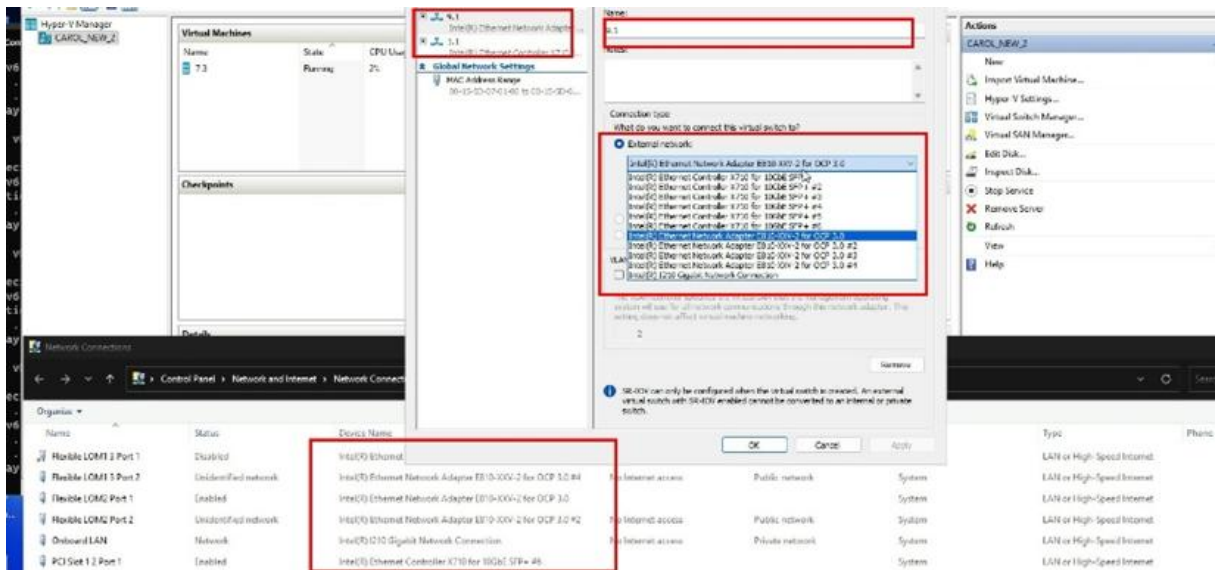


12. Select N210TP OCP with SR-IOV support, in connection type select “Enable single-root IO virtualization (SR-IOV)”. 設定 VM,可以設定很多組

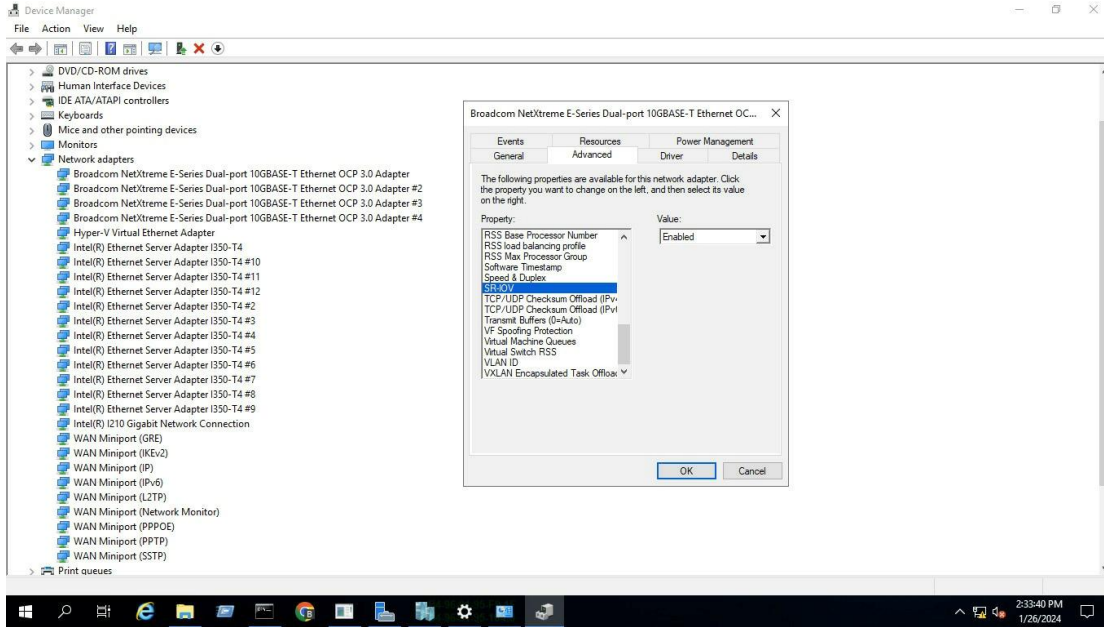


Name 自己設定

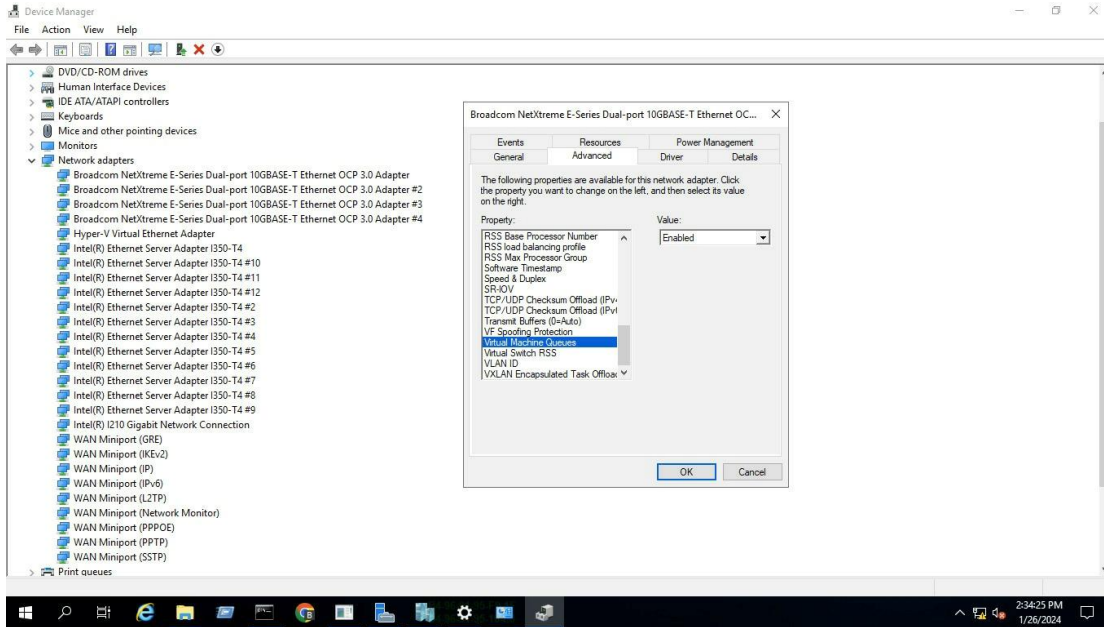
3 張 lan 卡就每一張要跑第一個 port 所以這裡就要設定 3 個



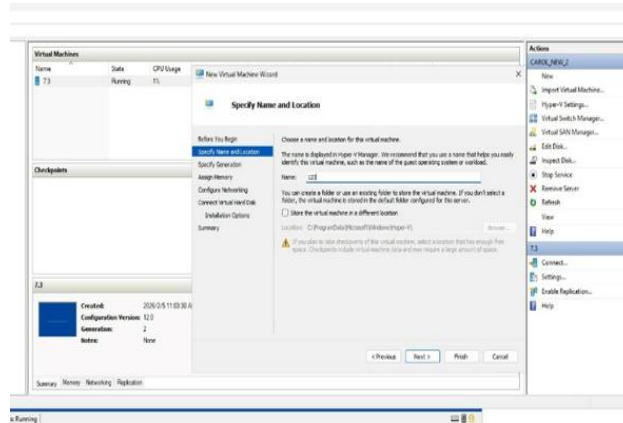
13. Open Device Manager / Advanced / SR-IOV enabled



14. Device Manager / Advanced / Virtual Machine Queues enabled



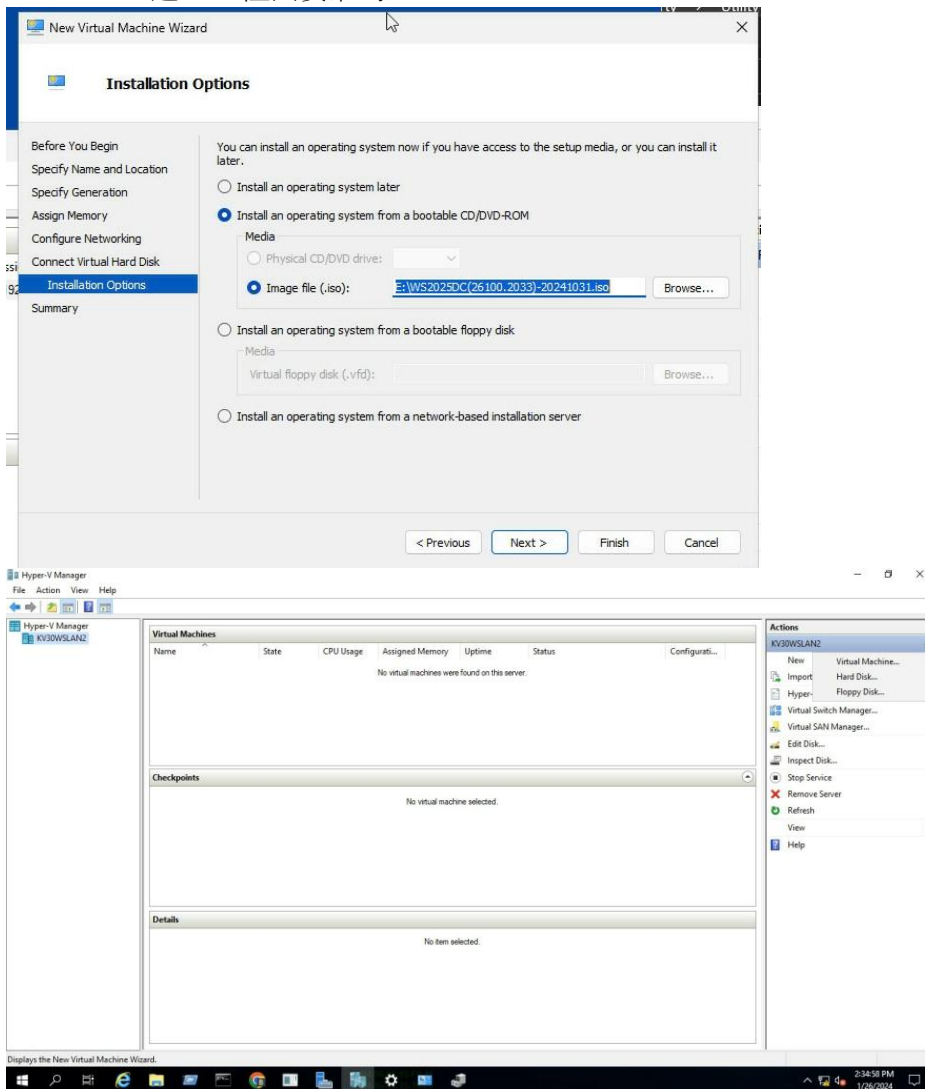
New > virtual machine(建 VM 命名)



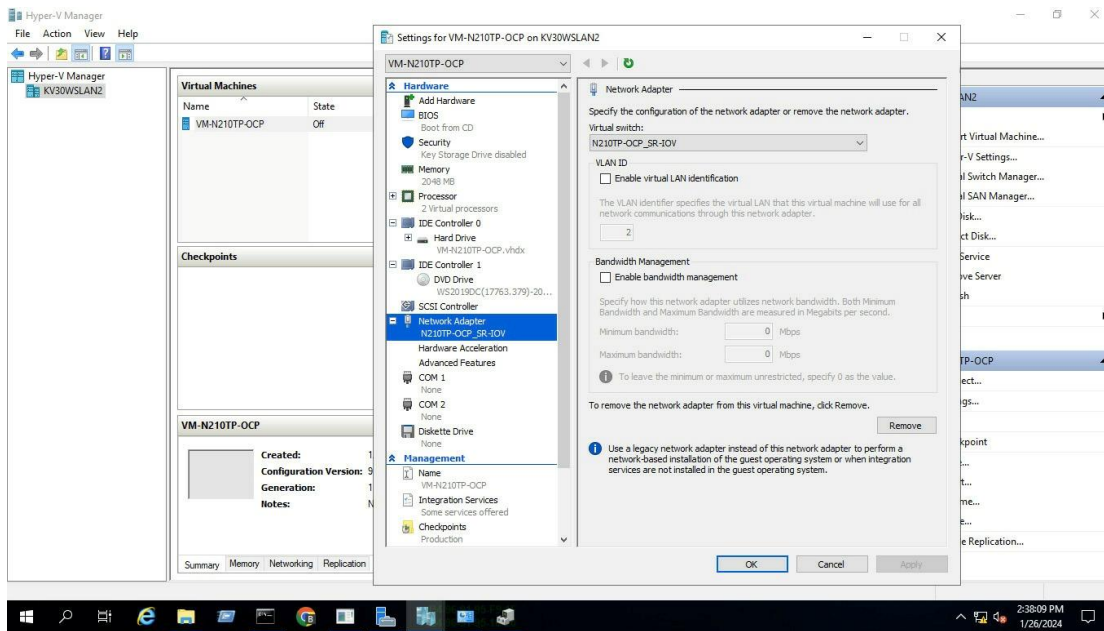
16. Assign memory :4096MB

17. Configure networking: 選剛剛建的 lan

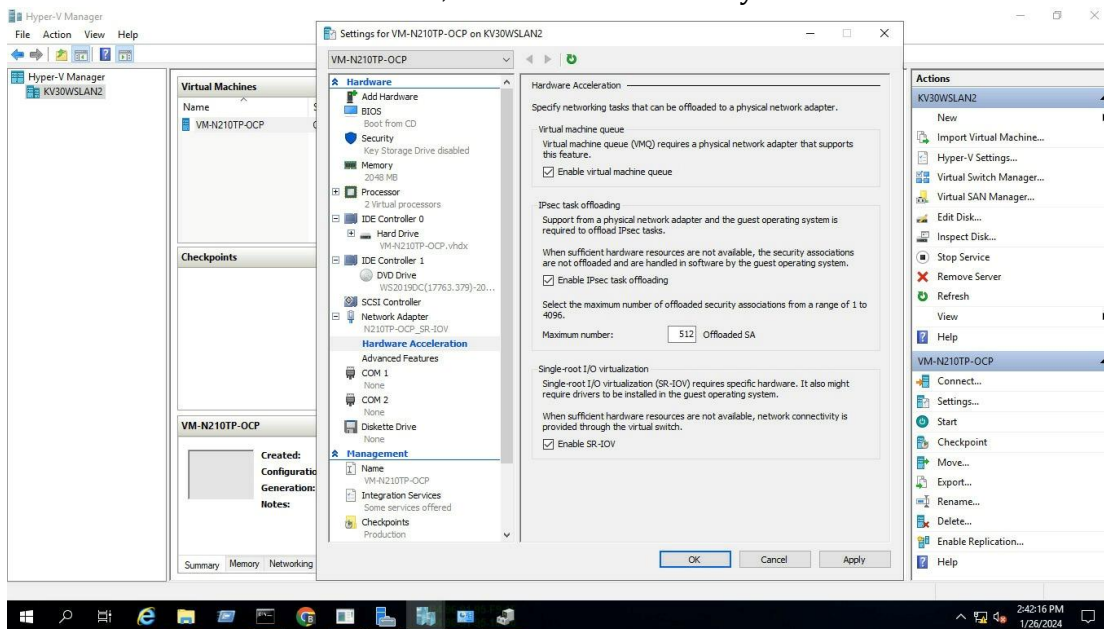
18. 選 usb 裡面要灌的 iso



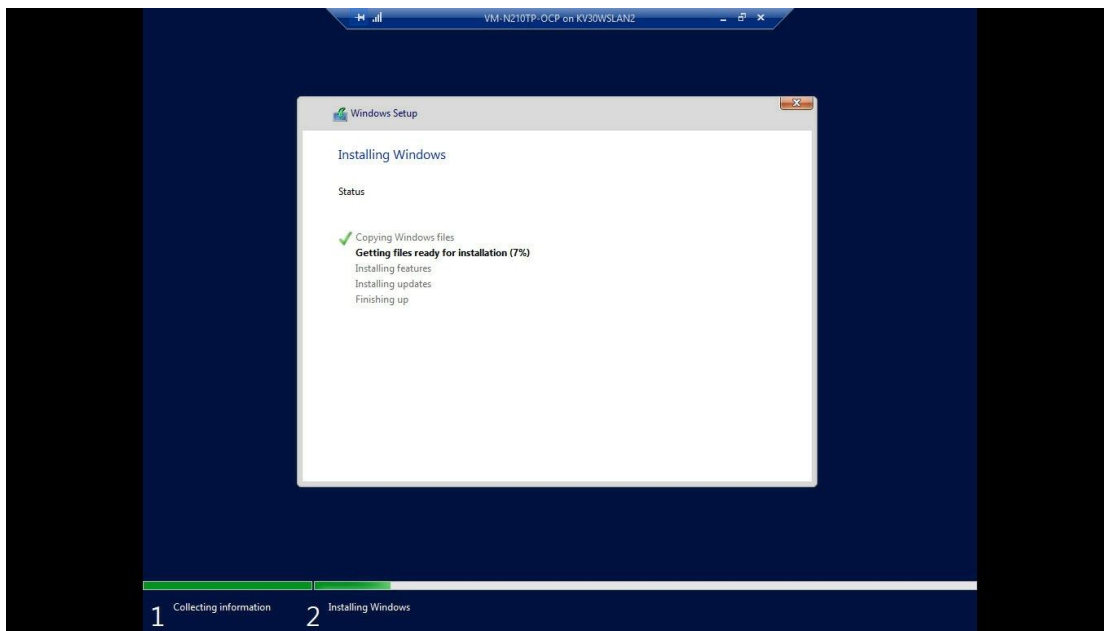
19. VM settings / Network Adapter. (右鍵選 setting)



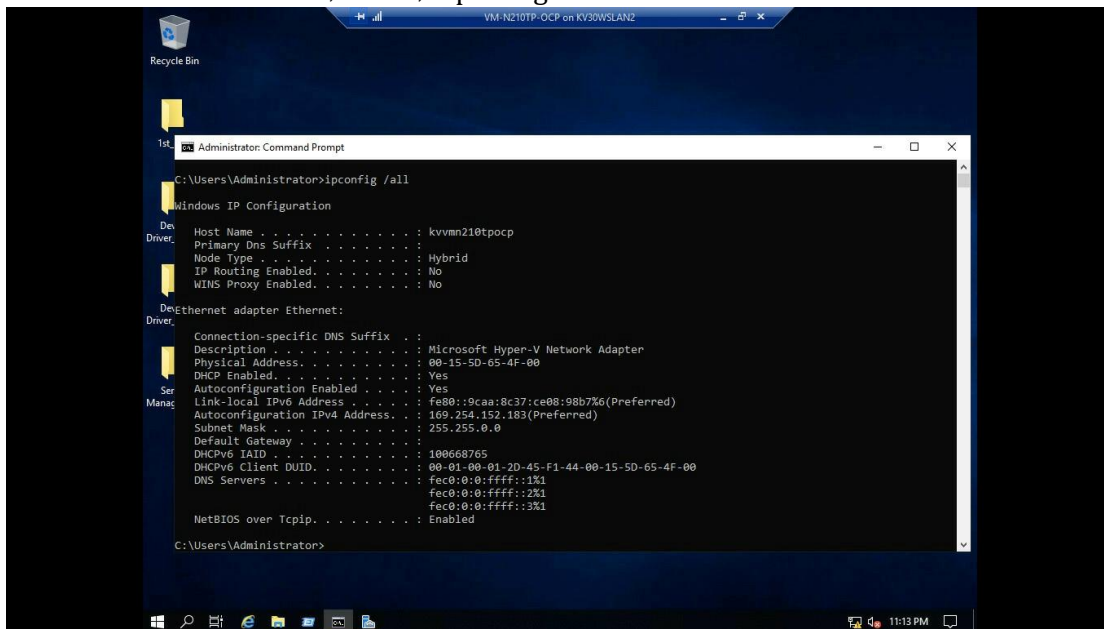
20. Hardware Acceleration / Enable SR-IOV. >APPLY



21. Installation VM OS. > 開啟 hyper V > state : connect

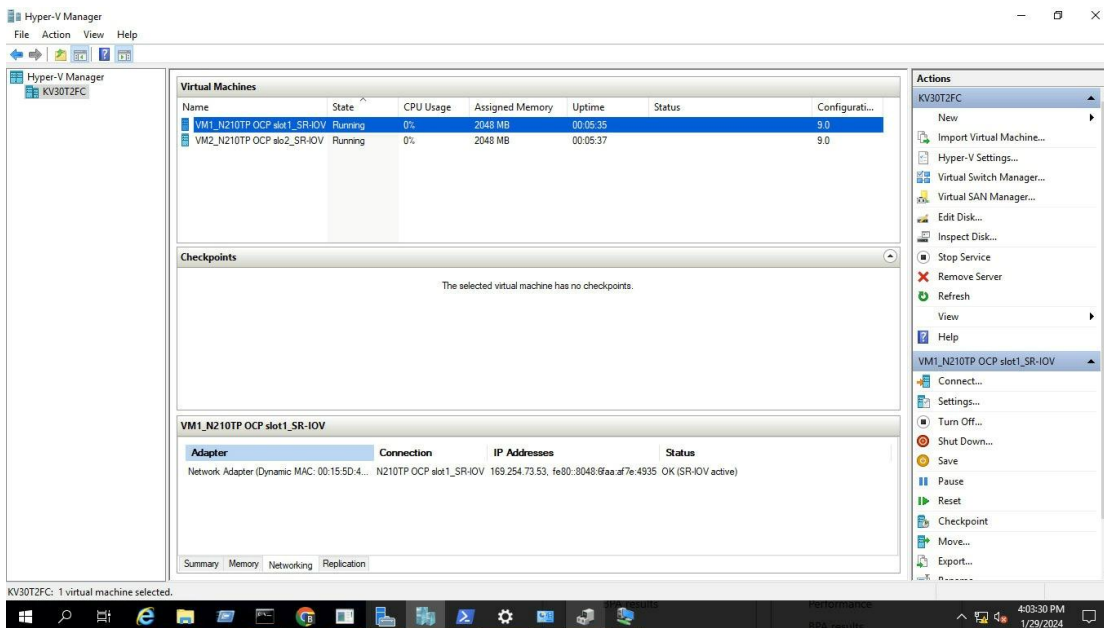


22. Boot in VM OS / CMD / ipconfig



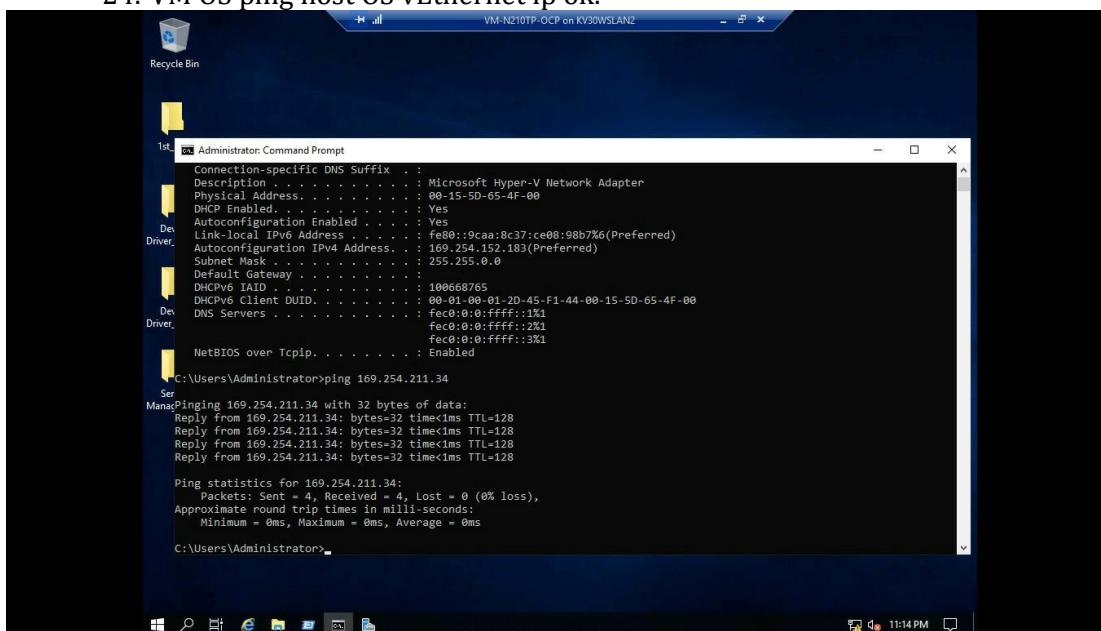
23. Hyper-V Manager / VM / networking / N210TP OCP status OK,
(SR-IOV active)

★ If the SR-IOV status is Degraded, try updating the drivers in the VM OS.

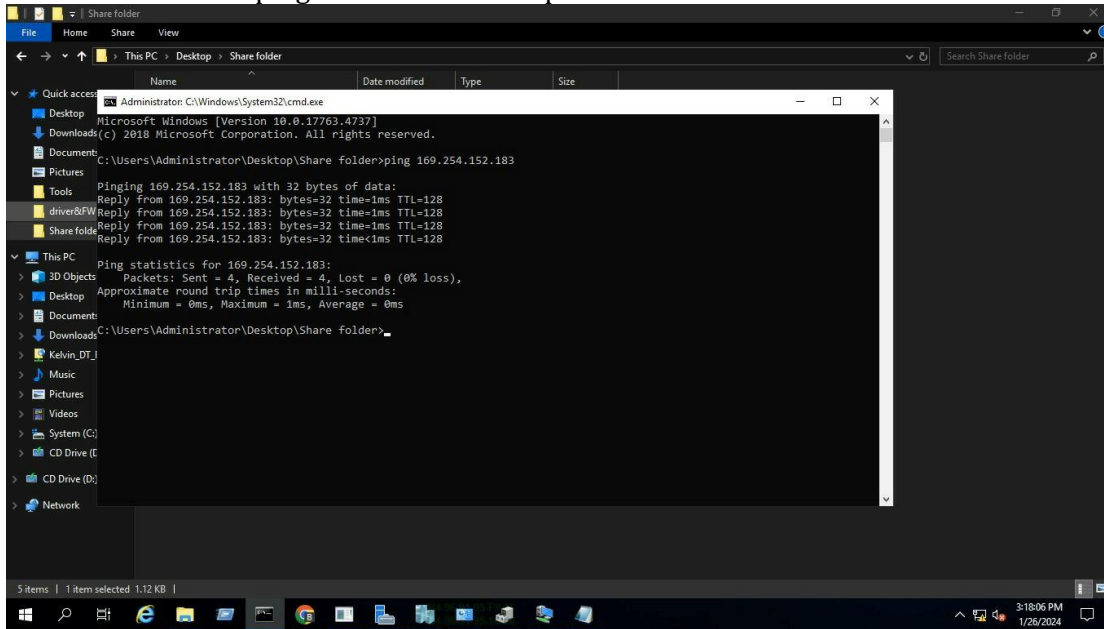


★ The following steps are to verify whether the SR-IOV function can be used.

24. VM OS ping host OS vEthernet ip ok.



25. Host OS ping VM OS vEthernet ip ok.



26. Run Powershell command : get-vm switch *fl*

IovEnabled = True

